Exercise 6 :

a)

$$E[Y_t] = E[R\cos(2\pi(ft+\phi))], \quad R \perp \phi$$
$$= E[R]\,E[\cos(2\pi(ft+\phi))]$$

$$E[\cos(2\pi(ft+\phi))] = \int_0^1 \cos(2\pi(ft+x))\,dx$$

$$= \left[\frac{1}{2\pi}\sin(2\pi(ft+x))\right]_0^1$$

$$= \frac{1}{2\pi}\left(\sin(2\pi ft + 2\pi) - \sin(2\pi ft)\right)$$

$$= 0$$

finally : $\boxed{E[Y_t] = 0}$

b) $\forall\ k \in \mathbb{N}$

$$\mathrm{cov}(Y_t, Y_{t-k}) = E[Y_t Y_{t-k}] - E[Y_t]E[Y_{t-k}]$$
$$= E[R^2\cos(2\pi(ft+\phi))\cos(2\pi(f(t-k)+\phi)] - \underbrace{E[R\cos(2\pi(ft+\phi))]}_{=0 \text{ cf (a)}}\underbrace{E[R\cos(2\pi(f(t-k)+\phi))]}_{=0 \text{ cf (a)}}$$

(trigonometry formula)